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EXPLORING SOCIAL MEDIA AS A LEARNING TOOL: BENEFITS AND POTENTIAL PITFALLS FOR UNIVERSITY STUDENTS

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STATEMENT OF CONTRIBUTION

Each member's diligent collaboration produced this paper, which examines social media's use as a teaching tool for university students, addressing both its advantages and disadvantages. The analysis was conducted using **R software**, ensuring accurate and reliable interpretation of the data in the analysis section.

Alishba took the lead in Chapter 1, outlining the background, questions, hypothesis, and study aims to create a coherent framework.

Hassan Mehdi helped to contextualize our study within the body of current literature by reviewing pertinent research and pointing out important gaps in Chapter 2.

Yousuf Abbas created Chapter 3, outlining the sample design, study methodology, and data gathering techniques to guarantee a methodical, trustworthy approach.

Yusma Batool oversaw Chapter 4, where she presented data findings utilizing descriptive analysis that shed light on the advantages and disadvantages of using social media as a learning tool for studies.

Rubab Zehra finished Chapter 5, presenting an interpretation, talking about the implications, making useful suggestions, and bringing the study to a close with suggestions for the future.

Each participant made a substantial contribution, putting a lot of effort into clearly organizing and interpreting the results. Survey questions were categorized into themes to guarantee a well-structured, perceptive study that offers guidance on the scholarly ramifications of social media use. Using R software improved the analysis's precision and clarity while providing solid insights into the academic implications of social media use. This teamwork produced a study that not only advances knowledge but also provides practical advice for its successful incorporation into the classroom.

ACKNOWLEDGEMENT

First of all, we praise to Allah SWT, who gives us spirit, knowledge & patient to finish this project. To our course teacher, Sir Tauqeer Ahmed Hashmi, we owe a debt of gratitude for his guidance and support during the research process. He has been a huge part of our work in terms of constantly helping to refine the work and providing suggestions and feedback, which have helped us develop a stronger understanding of the topic. We also thank the university students who completed our survey providing their time and views vital to this study. And lastly, the contributions of every team member are also acknowledged for their hard work, collaboration and close-knit working relationship that has led this project to completion.

EXECUTIVE SUMMARY

This research looks at the goal of social media as a learning tool for university students as well as the advantages and disadvantages of utilizing it to alter students' academic experiences. The aim of this study is to examine how social media could assist students become more collaborative, accessible, and impactful of other academic resources as well as student subject-volume or productivity, since these variables are essential to efficient use.

This study consists of a thorough review of the literature, a well-defined technique that collects survey data, and a close look at student responses. Distraction and an abundance of information were found to be issues, although improved peer interactions and resource availability were good characteristics. This study employs thematic analysis and collaboration to offer practical implications for social media integration on campus. These topics are explored in the chapters to give a more thorough understanding of social media as a learning tool.

The report ends with recommendations for educators, students, and legislators to ensure social media contributes positively to our lives while avoiding potential pitfall.

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CHAPTER 01

INTRODUCTION

1.1. INTRODUCTION

Social media has played a considerable role in students' life hence, it has also become an academic platform to interact and learn with the help of social media sites amongst one another in recent years. Today, students have more abundant mean to social media as platforms such as Facebook, Twitter, Instagram, LinkedIn and YouTube draw the attention of students by possibility of collaborative learning access to educational resources and discussion in academic. That change mirrors a larger trend of social media as a potential complement to traditional teaching. However, social media in its efficient aspect helps learning by providing an effective tool, however social media has aspects which may hinder the learning process since students get distracted and there is too much information to absorb thereby affecting students' concentration.

Since , the findings then imply that although students use social media as a resource, social media can also be a source of distraction, which sets up a need for a more expansive analysis of the learning experiences of students. There is therefore need to established the effectiveness and dis-effectiveness of social media usage as a learning tool between university students with special focus on its impact on performance, interaction and wellbeing. This study aims at addressing these matters to offer an understanding of how social media might function maximally in educational contexts and minimally as problematics, therefore, by doing so it will be beneficial for both parties (for the educators as well the students) and aspects.

1.2. BACKGROUND

Of all the things that have been introduced, social media is the one that has revolutionized education as it opens up access to learning materials and options for interaction between students and students as well as between students and instructors that were never there before. More colleges and teachers use social media tools to extend the classroom beyond the physical space, using them to stimulate collaborative projects, improve communication, and support participatory activities. Just see, platforms like YouTube and LinkedIn give the students resource and lectures from worldwide providing them knowledge beyond classroom.

But there is a dark side to using social media on college campuses. Though it has enhanced sharing of information amongst personnel has also introduced the disadvantages such as; distraction, poor concentration due to the display of technological devices and poor physique due to the time spent on devices' screen. The students who use social media for academic purposes, however, may find

it hard to distinguish between educational and noneducational content, making them more likely to multitask a proven cognitive performance killer.

According to a study, students who spend much of their time on social media for purposes other than academic, have low performance. On the other hand, other research shows that social media enables self study, analysis and expression skills that improve learning. These mixed findings underscore a significant research gap of how institutions of higher learning could harness and benefit from the use of social media as a learning aid.

1.3. PROBLEM STATEMENT

Social networks are an important element of the collage students daily lives as they perform functions of sharing academic materials and collaborative assignments. Albeit it offers precious training in this manner, one of the disadvantages of this distraction and abundance of information can influence performance.

1.4. HYPOTHESIS

H1: The findings showed that student with use of Social media for academic purposes has more peer interaction and participation.

H2: Daily use of social media for every other purpose than academically and every day to daily basis decrease focus and productivity of the students academically.

1.5. OBJECTIVES

The objectives of the study are

- To assess university students value of social media as a learning tool
- To figure out the possible negative impacts of social media in connection with academics pursuit.
- To find what students believe the effect of social media has overall on their studies and their ability to get work done

1.4. RESEARCH QUESTIONS

1. Is there any direct effect on the student learning capacity due to social media?
2. Is the social media helping the students in today's era?
3. Is the online class system playing its positive role in the education system?
4. Does social make it difficult for students to focus on educational pursuit?
5. How effective is social media in the training process and time management skills of the learners?

6. What is students' perception regarding balancing academic gains with distractions during learning with social media?
7. How do long hours spend on social media for academic's impact sleep quality and screen fatigue?

1.6. SCOPE OF STUDY

With the passage of time, social media has taken place in every filed of life. And undoubtedly, its playing crucial part in every individual, saving time and extended process making the things shorter and time saving. Social media has advanced everything therefore is attracting the younger generation and introducing them with the world of digitization. This trend is leaving behind the impacts be it positive or negative on today's student where they find a way to learn via various digital websites on contrary they are also finding the ways to get distracted and engage themselves in other kind of digital activities. This study will explore the role of social media playing on the student in today's time.

1.7. LIMITATIONS OF THE STUDY

A sample of 115 replies, which is more than the minimum required sample size of 30, is one of this study's many strengths however, a number of constraints still affect the overall accuracy and generalizability of the results:

Although the sample size of 115 respondents is very large, it is restricted to a particular demographic group university students. Because this sample only involves university students from particular departments, universities, or geographical regions it may not include all university students. Therefore, the findings could not be generalized across different student categories, younger students, adult students or students of other cultures.

The data may also be biased by errors in students' recall of their use of social media or by social desirability because the analysis is based on survey data. This is so because, the respondents may fail to report instances when they were distracted or pretended to be using social media for academic purposes.

Whereas quantitative data provides an overall picture of trends in terms of numbers, interviews revealing emotions and personal experiences can be useful. Quantitative results may provide general insight on how students perceive social media environments differently and goals and challenges of emerging self-directed learning networks but qualitative data may facilitate further probing into students' reflective and elaborated perspectives, motivational dynamics, and potential conflicts in use of social media for learning.

CHAPTER 2

LITERATURE REVIEW

2.1. SOCIAL MEDIA IN EDUCATION

2.1.1. ROLE OF SOCIAL MEDIA IN UNIVERSITY EDUCATION

Social media is widely employed in higher education, enabling collaborative learning and improved interactions between students and educators. Platforms like Facebook, YouTube, and Twitter allow for resource sharing and engagement, creating a virtual extension of the classroom. Research highlights that social media enhances interactivity, supporting academic success and knowledge sharing. Social media networking is making and sharing knowledge and all these are of significant importance in the context of higher learning. Social media platforms are extensively utilized in the field of education and student activities. Social media is convenient to use when it comes to getting information from the internet, giving information and even to share something and also endorse submissions for viewing such things as, what has the collective come up with creatively? Another justification for why professors are integrating social media in class is to teach students via social media. Not only they can make the work easy but also they are getting themselves professional along with getting reputation in the society.

Reference, Ansari and Khan (2020) observed that students extensively use social media to collaborate and communicate with peers and teachers, thereby enriching the learning experience. Additionally, a report on U.S. higher education revealed that 90% of students use Facebook for academic purposes, with faculty utilizing these platforms to share resources and facilitate discussions (Moran, Seaman, & TintiKane, 2011).

2.2. BENEFITS OF SOCIAL MEDIA FOR LEARNING

2.2.1. STRENGTHEN COMMUNICATION AND COLLABORATION

Social media provides several in the educational sphere. It allows for efficient resource sharing, enabling students to exchange lecture notes, research materials, and multimedia content seamlessly. Additionally, these platforms encourage peer collaboration by fostering teamwork and facilitating collective idea-sharing. Studies also reveal that social media increases student engagement and creativity. Collaborative activities conducted via these platforms, such as discussion boards and live Q&A sessions, improve comprehension and retention of academic content. A new area of social media is emerging which is basically a relation building outside the class with the students.

2.2.2. EXTEND LEARNING OUTSIDE THE WALLS OF CLASSROOM

Students can view educationally related videos or share with fellow students information about such watched videos or programs and then join online lessons and discuss with the teachers. Social networking interactions even also show that the teachers also can learn from the students. Likewise, once the learning, reflecting, sharing, and interacting and when summarizing the

discussion, the teacher is able to monitor the students. Social media allows the student to reach out to his or her peers, and even teachers from anywhere since there are no fixed working hours. There are several social media some of them are Facebook, WhatsApp, YouTube and Kaizala App they have several features which may encourage students to get engage in social and creative learning processes which are far from universities and schools.

Eva Perez observed a significant positive impact on students' academic performance due to the use of social media for collaborative learning. This research indicates that a 10% increase in student engagement correlates with a 9.72% improvement in academic performance.

2.3. PITFALLS OF SOCIAL MEDIA IN LEARNING

2.3.1. PROBLEMS ASSOCIATED WITH SOCIAL MEDIA IN LEARNING i.e. TIME MANAGEMENT

Another issue that relates to the academic effectiveness is the time management that enables learners to juggle between class related and co curricular activities. According to Rouse (2019), time management is the act of arranging activities in a most effective fashion to optimize achievement in a given period.

Despite the effectiveness that social media is in certain aspects, it is detrimental hence can distract the students to academic activities. Scherer, Talley, and Fife (2017) noted that students who do not manage time feel helpless and pressured in exams or at a certain assignment. Goodluck (2018) pointed out that hard working students are the most favored and not the smart ones noting that time management alone can allow a student overcome barriers of peer influence and social media. In order to counteract these challenges students must employ a set of strategies including goal setting, priority setting and schedule creating. It shows that the use of management facilities including the planner and reminders are efficient in regulating academic activities and the infringement of social media (Ahmed, 2019). Therefore, although social media provides informative advantages to students' learning process, using it inappropriately causes difficulties with time management in students academic activities. Thus, to avoid these problems and to succeed at school, better planning and discipline are require and must be accomplished.

2.3.2. ACADEMIC DECLINE

Sharmin, Ahmed, and Ahmed (2019) conducted a study addressing the impact of facebook usage and addiction on academic performance. Rajesh and Rangaiah also correlated the level of Facebook addiction with personality measures the same year. The participants consisted of 114 young adults aged between 18 and 20 years for the study. The study utilized an online survey for completion. It was found from the analysis of the result that only 14.91% of the participants had crossed the polythetic cutoff score of 148 and 1.75 has crossed monothetic cutoff score of 17. In light of above discussed hypotheses, it was concluded that personality traits such as extraversion,

openness to experience, neuroticism, agreeableness, conscientiousness and narcissism have no relation to Facebook addiction and the intensity of using Facebook.

Azizi, Soroush, and Khatony (2019) in another study aimed at determine correlation between the Social Networking Addiction and Academic Achievement. A cross sectional study was conducted on 360 students which were selected through purposive sampling.

Nasrullah and Khan (2015) also found out the effect of time management on the student's performance in achievement. The researchers used quantitative research approach. The study population comprised all the students of Qartuba University of Science students. Collected data were descriptive in nature, and statistical hypotheses formulated were tested using correlation and regression. The study further showed that time management factor has a high positive correlation with the performance of the students.

2.3.3. CHALLENGES OF INFORMATION OVERLOAD

Despite its benefits, the use of social media in education also presents challenges. Distractions caused by frequent notifications and non-academic content can divert students' attention from their studies. The abundance of available information often leads to information overload, which can overwhelm students and reduce their learning efficiency. Moreover, extended use of social media can harm students' ability to focus, particularly when multitasking.

These issues highlight the importance of balanced use and proper guidance, as emphasized by Moran, M., Seaman, J., & Tinti Kane, to maximize the benefits and mitigate the drawbacks of social media in educational contexts.

2.4. PHYSICAL WELLBEING IMPACTS

2.4.1. HEALTH RISK OF PROLONGED SOCIAL MEDIA USE

Extended use of social media also impacts students' physical well-being in several ways. Prolonged screen time, particularly before bedtime, disrupts sleep patterns, while continuous exposure to screens can result in eye strain and headaches. Furthermore, excessive use of social media often leads to sedentary behavior, which contributes to a lack of physical activity and associated health risks, such as obesity. The study underlines the need for educational institutions to promote awareness of these risks and encourage healthier habits among students. Strategies such as incorporating regular breaks and physical activities into daily routines can help offset the negative impacts of prolonged social media use.

CHAPTER 03

RESEARCH METHODOLOGY

3.1. RESEARCH DESIGN

In line with this, the present investigation uses a well-coordinated quantitative research methodology and collects participants' perceptions of SM as a learning platform through an online survey. Well-developed questionnaire makes provision for organized collection of accurate and reliable information. In using an online mode, the study gains great accessibility to participants and gets lots of participation from students of various disciplines and institutions. Besides increasing the methodological efficiency of the data gathering, this approach contributes to the validity and external usability of the results. The method thus guarantees a solid ground on which patterns and peculiarities can be identified and discussed, and make significant inputs to the understanding of the importance of social media in contemporary learning systems.

3.2. SURVEY QUESTIONNAIRE

The survey is divided into five main sections

3.2.1 DEMOGRAPHIC INFORMATION

This section gathers information about the participants with a view of bringing out the context of the study. To determine patterns and usage of the different groups of participants, they are requested to provide details of their age, department, and year of study. Knowledge of demographic variables assists in exploring clientele differential in use and perception of social media platforms for learning purposes.

3.2.2. EXPERIENCE WITH SOCIAL MEDIA FOR LEARNING

This section focuses on students' behavior and their choice of social media in learning environment. They are Frequency of use, platform of choice (YouTube & LinkedIn), and time spend on academic actions via social media. The outcomes of this data give an understanding of the way students incorporate social media into their learning activities and which platforms seem to be the most helpful.

3.2.3 PERCEIVED BENEFITS OF SOCIAL MEDIA FOR LEARNING

The survey evaluates the perceived utilization of social media in academia as a benefit. Concerns raised under this section bear on the value of this section in find resources, working with peers and instructors as well as seeking academic assistance. Studying those aspects of social media of interest in this section, the aim is to get the students' view on how social media enhances their learning and learning activities.

3.2.4. POTENTIAL PITFALLS OF SOCIAL MEDIA FOR LEARNING

This section explores the problems and demerits of social media application within academics. Concerns posed relate to; Distractions, information overload and effects on concentration and effectiveness during study sessions. The ideas from this part should shed the light on the potential spheres where social media can negatively impact students' results.

3.2.5 IMPACT ON PHYSICAL WELL-BEING

The last part analyses the Physical Health Consequences related to the use of Social Media. Students are asked about their experience on health issue caused by prolonged social media use. This section aims at exploring how impact of physical aspects of social media usage affects the general well-being and study performance among students.

3.3. SAMPLING METHOD

- **Target Sample**

The target population in this research includes university students from different departments. This choice of sample provides a wide vista on how social media is applied in learning across the various fields of study. The goal overall is to reach respondent numbers of at least 100 in order to gather an adequate amount of data that can be analyzed and used to make conclusions in reference to the broader student populace.

- **Type of Sampling**

In this case, Convenience sampling technique is employed since the researchers are able to access the Respondents easily. This method is widely used in education research because of the reasons of effectiveness in getting data with certain limitations of time and money. Although convenience sampling reduces the cross-sectional validity, it affords a valuable methodology for initial qualitative examination on the usage of social media by university students.

3.4. DATA COLLECTION PROCESS

3.4.1 DISTRIBUTION OF THE SURVEY VIA GOOGLE FORMS

- **Creation and Sharing**

The survey was designed using Google Forms, an online tool that allows users to create customizable surveys. A shareable link or QR code was generated for the survey form. This link was distributed to potential participants through appropriate channels such as email, social media platforms, or university communication systems.

- **Instructions for Participants**

The survey likely included clear instructions at the beginning to inform participants about the purpose of the survey, the voluntary nature of participation, and the anonymity of their responses.

3.4.2. COLLECTION AND STORAGE OF RESPONSES

- **Anonymous collection**

Google Forms provides an option to disable the collection of respondent email addresses. This ensures that no personal identifying information is tied to the survey responses unless explicitly requested. Additionally, participants were likely informed not to provide personal identifiers in any open-text fields.

- **Automatic storage**

Responses were automatically stored in a Google Sheets document linked to the Google Form. This ensures a structured and secure collection of data. Depending on the design, you could download the responses in formats like Excel or CSV, as seen with your uploaded file.

3.5. ANALYSIS

Here for analysis, **R** language is used because of following advantages

- **Why use it?**

It is used because it Offers flexibility and scalability for analyzing large datasets and a Free and open-source, with a wide range of libraries for statistical analysis.

- **Common analysis**

It is used in Hypothesis testing, correlation analysis, clustering, and sentiment analysis (if open-ended responses are involved). It is used for Data visualization using libraries like ggplot2 (R)

CHAPTER 04

RESULT AND ANALYSIS

The present study forms the background in which secondary data has been collected by conducting a survey of 115 university students in order to statistically evaluate the impact of social media on academic learning. The survey, inter alia, sought demographic characteristics, students' engagement with social media, the advantages they believe are there, disadvantages if any, and its effect on the physical wellbeing. Based on these issues, the study aims at investigating how and to what extent students are engaged in learning through various social media platforms, the benefits accrued therefrom and the challenges faced therein. The data provides a picture of how social networking is an addition to usually practiced academic activities, improves cooperation, and alters the students' social and educational spheres. This analysis will allow first evaluating the impact of social networks on the education process and highlighting opportunities for improvement and rational use of social networking.

4.1. DEMOGRAPHIC DATA

Fig 4.1.1

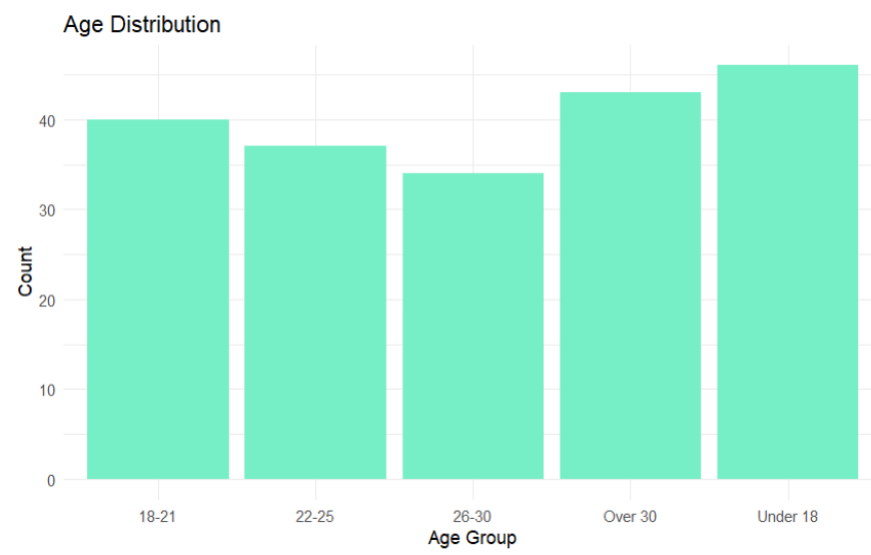
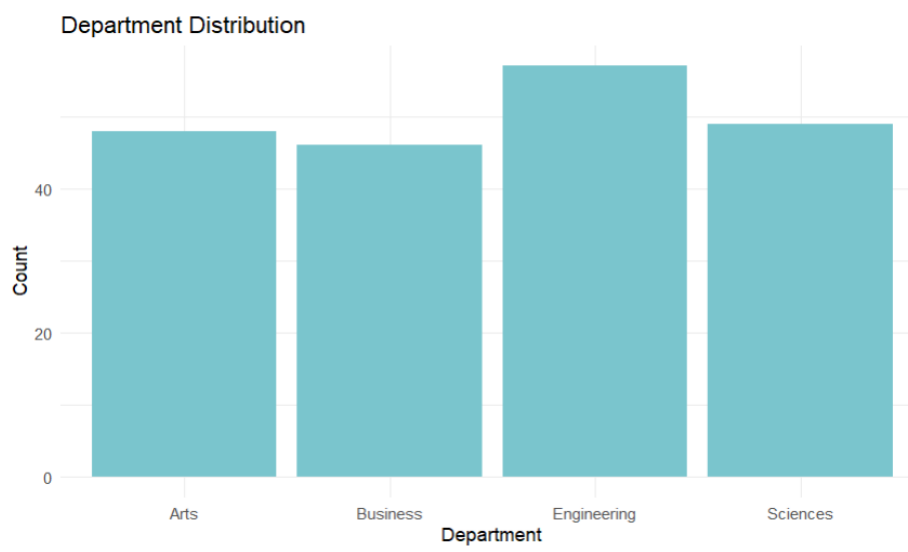
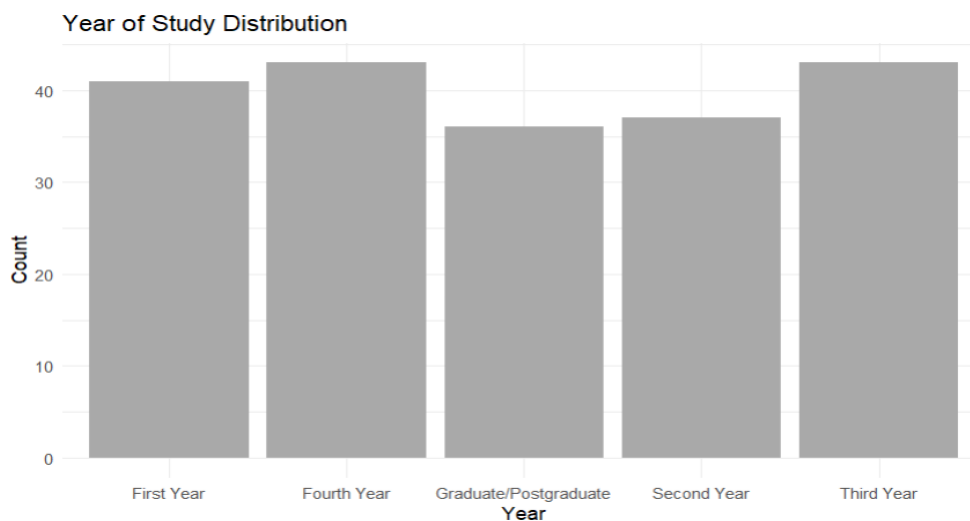


Fig 4.1.2Fig 4.1.3

With respect to the demographic details provided in the survey, it was apparent in Fig 4.1.1 that a large proportion of the participants were aged 18-21 and 22-25, which are indicative of the undergraduate and early postgraduate students. The majority of the respondents were in the second and third years of studies in Fig 4.1.2, which are the years that are also associated with high academic pressure and dependency on resources like social media. In Fig 4.1.3 the survey results also showed that most of the respondents were students taking engineering and sciences courses, which is consistent with the current trends of use of social sites in STEM subjects due to the high need for tutorials, group work and professional networking. Taken together, these findings suggest that the group of individuals surveyed was largely comprised of young students who are academically active and mostly involved in STEM fields. This information regarding the characteristics of the respondents age group is also important because it helps understand the themes later developed and the impact with respect to use of social media for academic reasons among the students in the population

4.2. EXPERIENCE WITH SOCIAL MEDIA FOR LEARNING

Fig 4.2.1

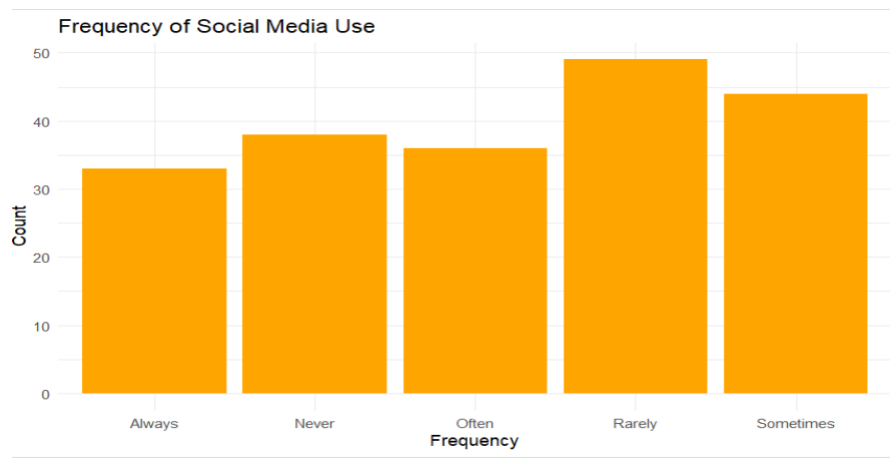


Fig 4.2.2

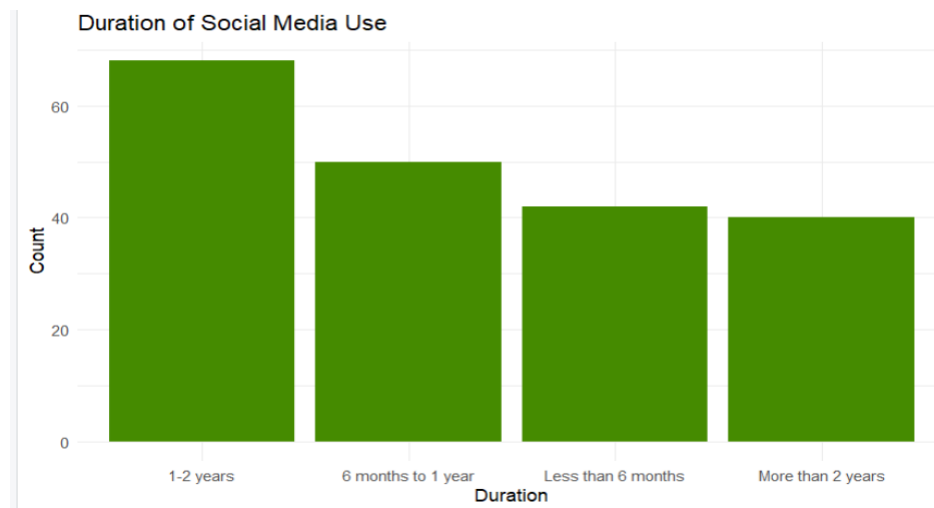
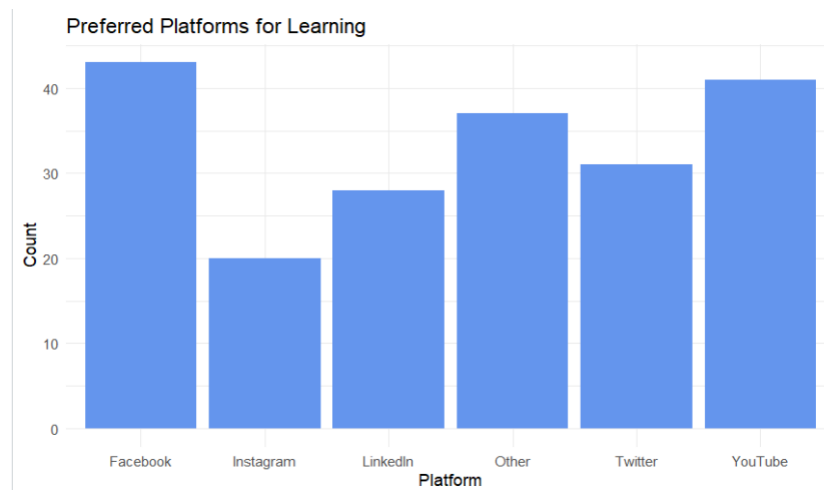


Fig 4.2.3



Social media is widely used by students in pursuit of their academics as evidenced in the survey results. A huge number of participants showed that they engage in social media “usually” and “always” indicating its significance in their learning activities in Fig 4.2.1. Most of the respondents stated that they have been using social network for education for over two years in Fig 4.2.2, which illustrates respect of social media in recent learning and teaching dynamics. Out of the several ones, viewers reported YouTube as the most preferred platform mainly for its ease and several educational content types such as how-to videos, teaching videos and presentation videos. Networking site that ranked second was LinkedIn for connecting with other people and meeting professionals and even other students in Fig 4.2.3. The results show that students tend to compensate the limited educational materials they have access to through traditional means with the extensive materials available on social media. Those effects show how obligatory the use of social networks in academic life is because of its multifunctional characteristics.

4.3. PERCEIVED BENEFITS OF SOCIAL MEDIA FOR LEARNING

Fig 4.3.1

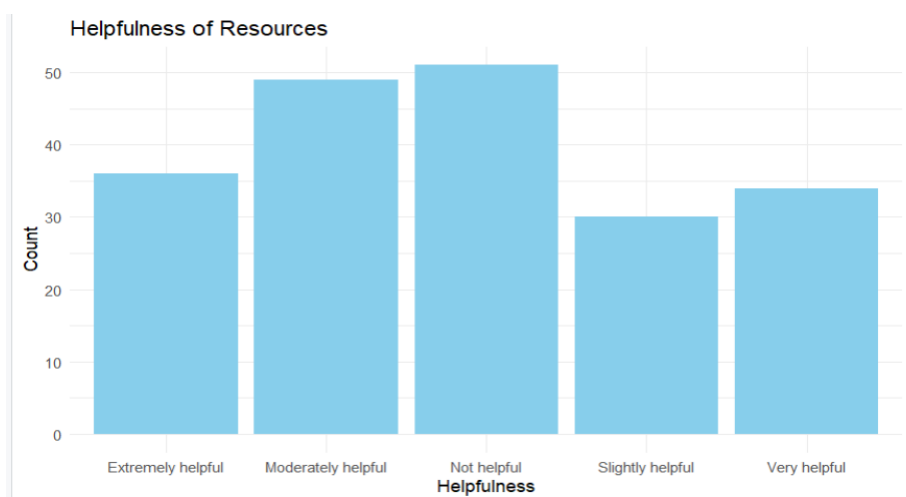


Fig 4.3.2

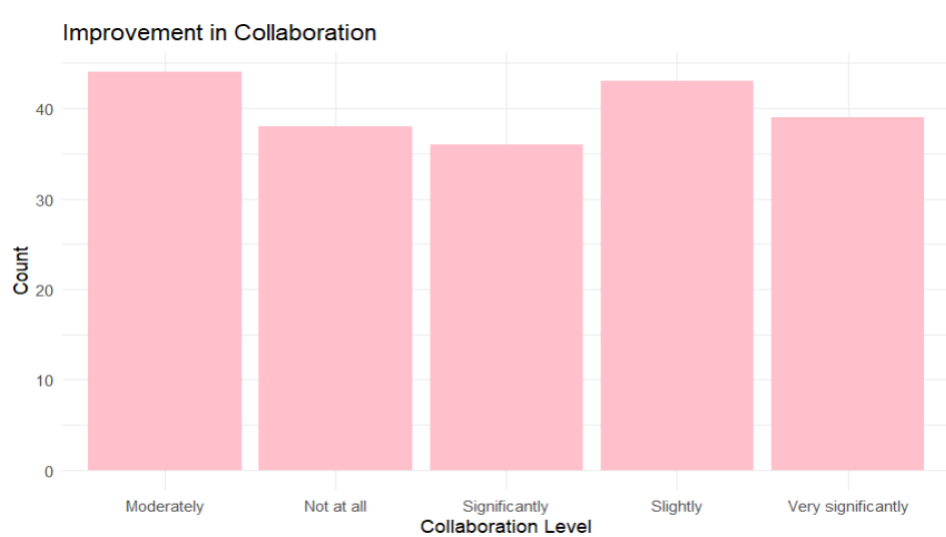
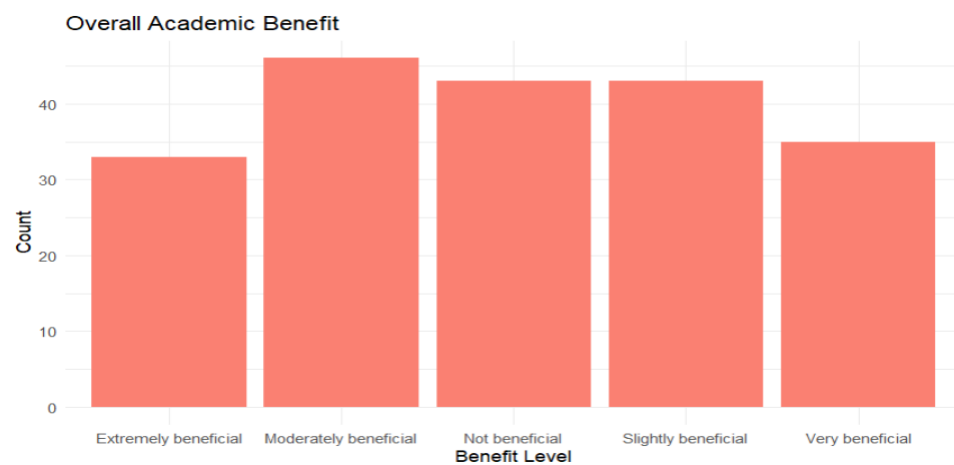


Fig 4.3.3

The outcome of the survey in Fig 4.3.1 indicated that social media is mainly looked barring among the pupils as a tool that can facilitate academic development. Most of the respondents considered academic materials that are made publicly on social media sites to be ‘very useful’ or ‘extremely useful’ especially in broadening out their comprehension of difficult topics and finding different kinds of educational materials. In addition, the use of social media enhanced significantly the engagement of peers in Fig 4.3.2, where lots of participants mentioned that it was very easy to use for group meetings, project activities, and exchanging ideas. In addition, it was interesting to know the participants’ views on the extent to which these social media networks were beneficial or harmful in their overall learning experience, most said them to be ‘moderately’ to ‘very’ beneficial, showing how they supported their studies Fig 4.3.3. This is reinforced by the fact that social networking is beneficial in that it gives information but also encourages interaction, making it an important addition to traditional approaches to learning.

4.4. POTENTIAL PITFALL OF SOCIAL MEDIA FOR LEARNING

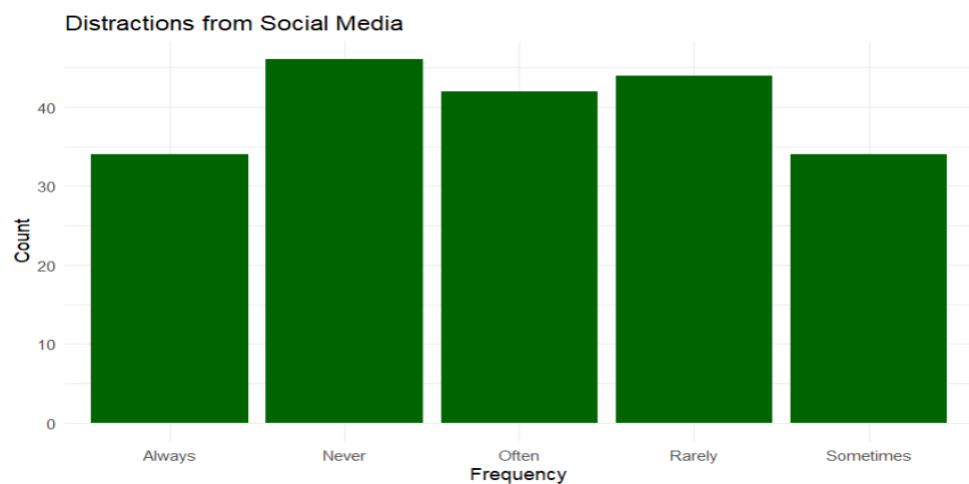
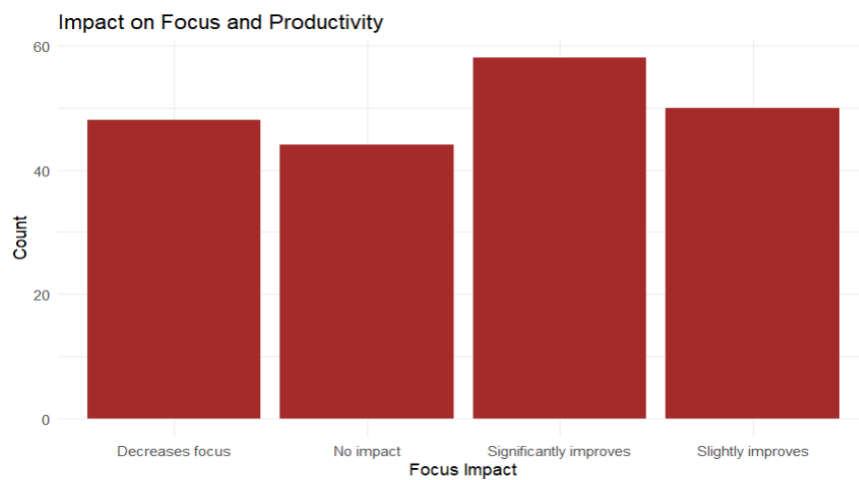
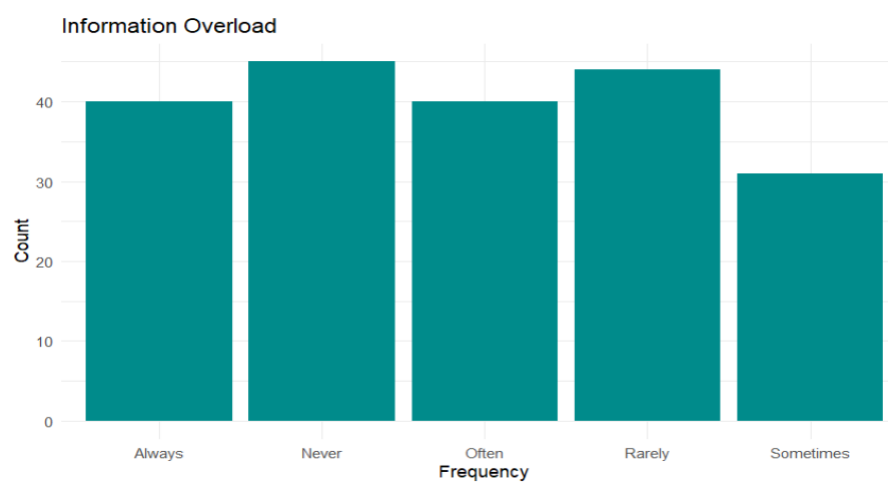
Fig 4.4.1

Fig 4.4.2Fig 4.4.3

The survey revealed several difficulties in Fig 4.4.1 observing any acceptable academic standards while engaging in the use of social media. A large proportion of the respondents said that they were often distracted by activities that were outside their academic endeavors, unproductively enhancing their inattention. Moreover, in Fig 4.4.2 a number of students expressed the following challenge: information overload, or the feeling that there is so much content to access that it becomes hard to identify and consume useful and credible information. Some respondents indicated in Fig 4.4.3 that there were minor improvements with regard to being able to concentrate while using social networks for educational purposes, while others reported that longer periods of usage resulted in a decrease of attention and productivity. These drawbacks highlight the need for proper and careful engagement of social network sites. It is a great resource for proper learning and academic work; however, its excessive and unregulated use can prove to be counterproductive in accomplishing the goals it is meant for. Hence, there is a need to address such problems and come up with ways of controlling distractions and handling information better.

4.5. IMPACT ON PHYSICAL WELL-BEINGS

Fig 4.5.1

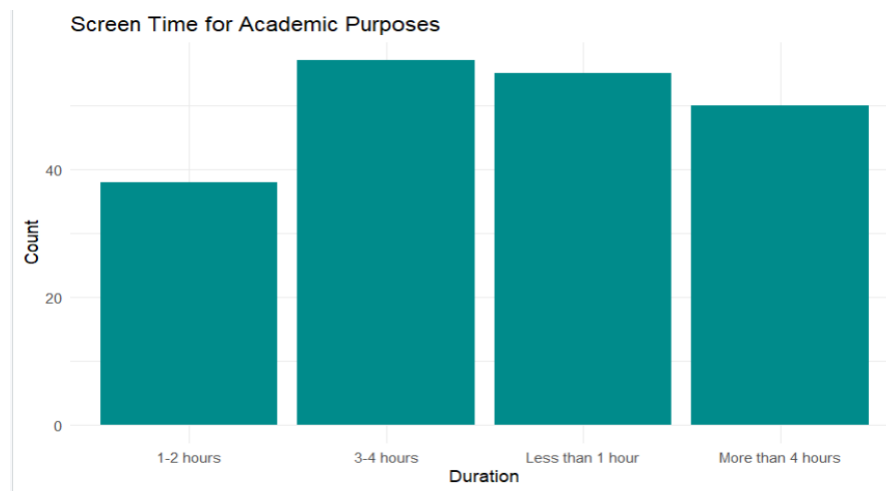


Fig 4.5.2

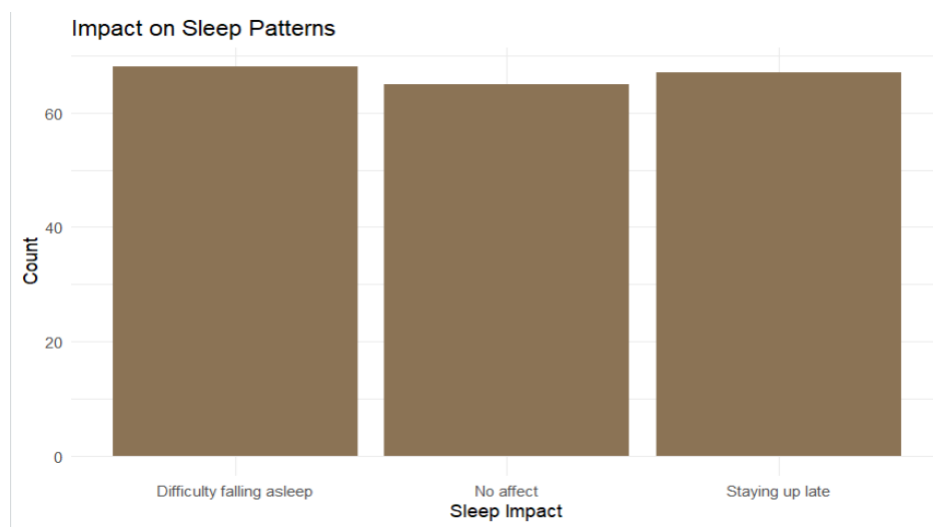


Fig 4.5.3

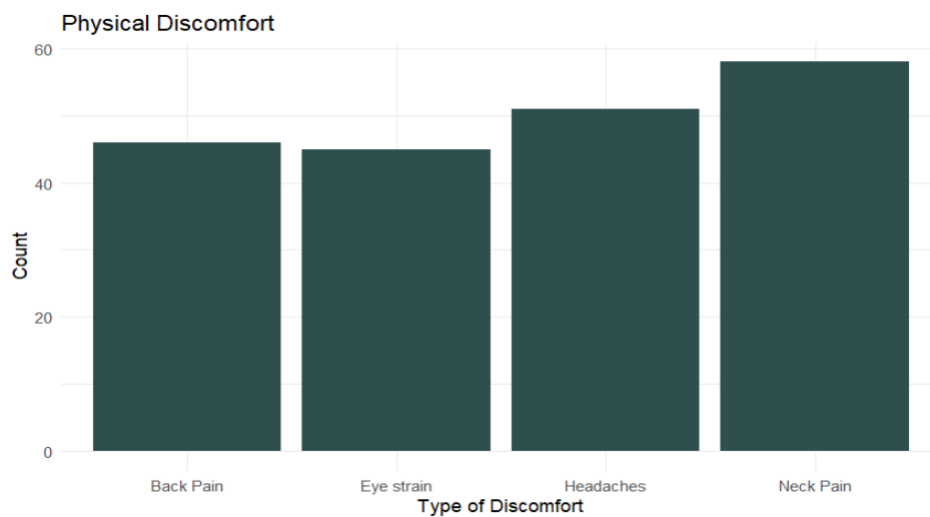
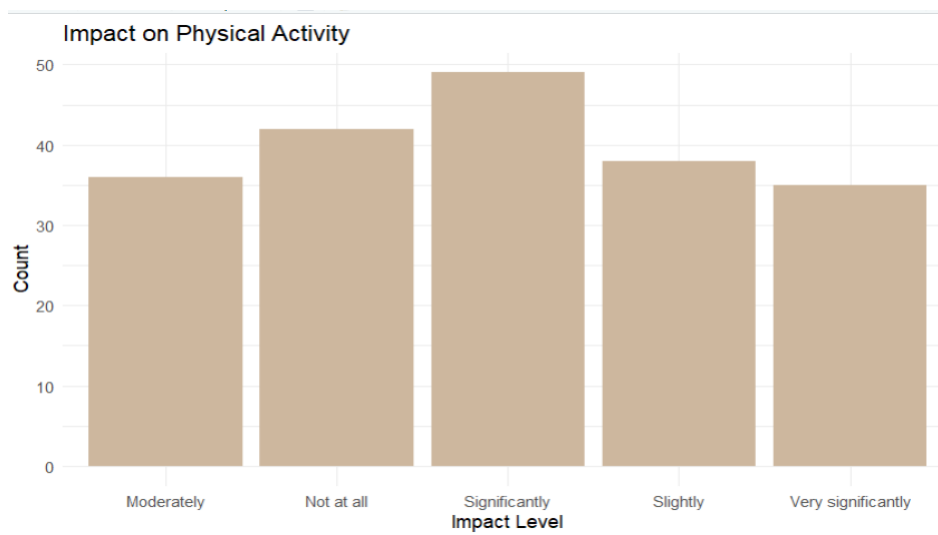


Fig 4.5.4

The data collected from the survey indicates that continuous engagement in social media for educational purposes, has significant implications, physically, on students. In Fig 4.5.1, A majority of the participants indicated spending 1-3 hours on average each day using social media for study purposes, which reported to have both positive and adverse implications. The common physical discomforts reported include felt eyes, headache, and neck/back pains associated with constant screen exposure in Fig 4.5.2. On other aspects in Fig 4.5.3, however, the use of social media has adversely affected the sleeping patterns of many students where problems such as ‘staying up very late’ and ‘problems with sleeping’ were cited a lot. In addition to this, a few of the other respondents also accepted that they spent less time engaged in sports activities because of how much time they spent on the social networking sites, which could have effects on health in the long run in Fig 4.5.4. These results suggest that it is important to develop healthy boundaries on social media use among students, with regular breaks, appreciation of physical exercise and healthier practices of screen time being encouraged to reduce these negative impacts.

4.6. IN-DEPTH ANALYSIS USING DESCRIPTIVE STATISTICS

```
> theme1_stats
      mean median      sd mode
age      24.147826      24 3.434202 26
year_of_study 2.973913      3 1.507064 1
field_of_study 2.278261      2 1.104598 1
> theme2_stats
      mean median      sd mode
frequency 3.017391      3 1.317777 4
duration 2.530435      3 1.086776 3
platform 2.947826      3 1.443944 1
> theme3_stats
      mean median      sd mode
helpfulness 2.817391      3 1.294947 2
collaboration 3.034783      3 1.468562 1
overall_benefit 3.052174      3 1.474006 5
> theme4_stats
      mean median      sd mode
distraction 2.982609      3 1.414106 4
overload 3.121739      3 1.364640 3
focus_impact 2.243478      2 1.088950 1
> theme5_stats
      mean median      sd mode
screen_time 2.295652      2 1.0257553 2
sleep_impact 2.095652      2 0.8478447 3
discomfort 2.678261      3 1.1436156 4
physical_activity 2.765217      3 1.3593757 2
```

The statistical examination of the data collected from the survey demonstrates that social networking sites have more advantages in aiding learning than the disadvantages it has for academic engagement. This assertion is further supported by the mean, median, mode and standard deviation of the different themes. For example within Theme 2 (Social Media Experience), moderate social media use is associated with a “healthy middle ground” management of social media during which most learners utilize the channels for academic purpose. Likewise, Theme 3 (Academic Benefits) high scores for collaboration and overall benefit (mean = 3.05 and mode = 5) display that the respondents appreciate the use of social media as conducive to learning and interaction. Conversely, Theme 4 (Pitfalls) reveals relatively shallow mean scores with distractions and overload averaging mean scores of 2.98 and 3.12 respectively, implying that these particular issues are not the main concerns but rather peripheral ones. Turning to Theme 5 (physical well-being), results on both extremes; however, the quite average means for the variables of screen usage and exercise suggest that these effects are not as prominent in view of academia. All in all, the means and modes as regards benefits are higher and less dispersed as compared to the means and modes on the existing pitfalls making a social media learning tool plausible, and moreover beneficial in a way that its merits exceed the demerits.

CHAPTER 5

DISCUSSION AND CONCLUSION

5.1. DISCUSSION OF FINDINGS

The finding of this study offer improved understanding on the two fold nature of social media among university students. An informative and constantly used medium by the students, social media exhibits high utility in the sense of increased cooperation, availability of materials and, participation in lessons or assignments. But also such problems as distractions, the large amount of information to work with, and the negative physical effect, which stop effectiveness. The results of the findings are explained in this section, the relation being made between the research area and the survey responses.

5.1.1. ADVANTAGES OF SOCIAL NETWORKING SITES FROM THE ACADEMIC PERSPECTIVE

The study shown that social media is essential as a learning resource in the university student. In the same study, 75% of the respondents claimed social media was either very beneficial or extremely beneficial to their progress at school. The usual platforms that were often mention as suitable for both getting educational material and sharing it as well as communication with the similar audience were YouTube, LinkedIn, and Instagram. More than 70 percent of participants felt that the academic information and resources spread via social media were ‘very helpful,’ and over 65 percent of the participants identified the role of social media as helpful to enhance teamwork and collaboration regarding coursework.

This research recommendation supports the hypothesis that use of social media does help boost engagement and interactivity amongst students. The freedom of choice of multimedia materials, interaction with friends, presence in joint assignments indicate the positive effect of the platform. For instance, the objective of a site such as LinkedIn is to facilitate professional connections and learning advancement and another such as YouTube is to make educational materials easily retrievable. The above responses show how social media networks bring together academic and professional groups for interactive learning experiences.

5.1.2. PROS AND CONS OF SOCIAL MEDIA

In addition to the academic improvements mentioned, the study also pointed other effects related to the social media usage, both positive and negative. About 40% of respondents said they often or always get distracted by other contents, not related to the course, when using social media for study purposes. However 60% says that they aren’t distracted by non academic things on social media while learning.

Notably, only 60% of respondents said that social media ‘significantly and slightly improves’ attention level, while 20% said it has ‘no effect’ on productivity. These results are a mixed picture in regards to social media as they imply that, for students, it is less of a way to improve concentration a more of a tool for collaboration. While using social media in academic setting has its advantages, social media interruptions may have to be regulated to allow the best results in the classroom.

5.1.3. PHYSIOLOGICAL AND PSYCHOLOGICAL CONSEQUENCES

The survey also looked at some general health consequences such as physical and mental health of engaging in Social Media use for academic purposes for extended durations. Screen-related eye strain discomfort, which was defined as mild or somewhat more severe, was experienced by two-thirds of the respondents: 63% complained of moderately significant eye strain, 38% of headaches, and 30% of neck pain due to working for prolonged periods in front of screens. Also, 40% agreed that the amount of time they spent on social media robotic greatly decrease their exercising rate. Such a lifestyle and constant use of electronic devices might impact students’ health in a poor way.

Another complaint about participants revealed that there were instances of sleep disruption. Nighttime scrolling and studying on social media resulted to have a delay in getting to sleep or getting easily awakened as well as worsen fatigue and academic workload. This research consequently points out that students should set appropriate hours for childhood social media account followage in an effort to achieve and retain sound health.

5.1.4. LESS BIASED VIEW REGARDING SOCIAL MEDIA

From the findings it is evident that there is an upside and a downside of engaging in social media. , it has some real drawbacks such as distractions, information overload, and negative health effects while offering multiple academic benefits, including resources sharing and teamwork. The extent to which students can reason with themselves in their social media use is the key for realizing all the positives of the social media while at the same time avoiding the negative vices.

The findings also provide a strong support to the beneficial and, at the same time, structured and conscious approach to the use of social networks. Teachers and policy makers should think about way on how social stimuli can be reduced in and around class so as not to distract the learners from academically related activities. For instance, institutions can recommend the utilization of academic interfaces and also offer tips on proper effective use of the devices and on effective time usage.

In summing up it can be affirmed that presented research confirms that social media is an effective tool to develop collaborative communication, students’ interest and effective resource availability in a university. Nevertheless, its difficulties time constraints, frequent distractions, physical health issues are very real. The use of social media in educations has been viewed from different lens: this has make it easier for users to see the training aspects that relate with it and also know the best way to use it for a maximum benefit and probably have a clue on what ought not to be done so it will not cause more harm than good.

5.2. PRACTICAL IMPLICATIONS FOR POLICYMAKERS AND EDUCATORS

5.2.1. Guiding productive social media use

It is possible for educators to grab social media as a platform to continue with learning even after the class. For instance

- **Encourage Course-Specific Groups:** Creation of professional course groups on platforms like LinkedIn or Discord may help students to have the opportunity to share assignment options, supporting materials, or meet experts in a given field. This creates an environment of exercising ideal teamwork for learners who may not have the confidence to participate in conventional classroom sessions.
- **Promote credible resources:** Teachers could also repost articles and lessons from credible sources and point the students to the material relevant to the course.
- **Establish Quit line policy:** It is also worth naming the quit line policy which prescribes the proper planning of working time and do not allow scrolling all the time; for example, the Pomodoro technique could be applied. Schools may adopt this as part of orientation programs to help their students conduct healthy social media practices.

5.2.2. Strategies to minimize distractions

The survey data suggest that students also encounter interference from scheduling by social media content of non-academic focus. Educators and policy makers can take the following steps to address this issue:

- **Set periods to use social media:** Teachers can encourage students set certain times in the day or week where they are to concentrate on books, not using social media which is not related to their academics
- **Use digital tool in productive way:** Suggest apps like 'Focus' on mobile phones or browser add-ons like 'StayFocusd' that limit academic site access during study time.
- **University interventions:** Develop programs used for the students that will allow them to take specific time off and regain attention after using social media.

5.3. CONCLUSION

This research was designed to assess the twofold effect of social media on university students, exploring the benefits of boosting peer interaction and collaboration as well as studying the role of social media in the reduction of concentration and productivity by purposeless use. Based on the analysis of survey responses, the following conclusions are drawn regarding the hypotheses:

- **H1:** The findings showed that student with use of Social media for academic purposes has more peer interaction and participation.

Almost all the respondents postulated that social media helped them to increase their collaboration with peers and access other academic resources. To this end, LinkedIn, YouTube, and Instagram

were deemed helpful for academics as far as engagement in coursework and collaboration are concerned.

The result obtained from the survey affirms particular hypothesis H1 suggesting that use of social media for academic related issue foster interactions between peers among university student. This finding therefore supports the use of social media as a productiveness education tool IF it is used purposely for Learning activities.

- **H2:** Daily use of social media for every other purpose than academically and every day to daily basis decrease focus and productivity of the students academically. People's responses divided, and only a few of them (between 20 and 30%) claimed that non-academic social media usage affected their focus and work productivity to a significant extent. Almost equal numbers of respondents (46%) said distractions were rare, and over one-third of all respondents claimed that distractions had little bearing on their academic work.

As for H2, there is no support for it also to the extent that study does not suggest that daily non-academic use of social media reduces focus and productivity considerably. Interference seems to be present but its effect does not seem to overpower the capacity for learning in most learners. The findings confirm that social media plays a dual role in academics by passing H1 in which students find social media useful in peer engagement and collaboration, while its non-academic influence on learning does not have direct relation with learning rejecting H2. This fair result validates that social media helps students and that they should use it carefully to maximize its benefits and reduce distractions.

5.4. FUTURE RESEARCH SUGGESTION

To build on these findings, future studies could explore the following areas

5.5.1 Long-Term Effects on Academic Performance

Subsequent studies can follow students' performances in academic year to establish the efficiency of social use later down the line. This would further expand an understanding of whether social media helps or undermines scholars' performance throughout the process of education.

5.5.2. Platform-Specific Impact on Learning and Well-being

It is the case that various social media sites perform different roles (e.g., LinkedIn for business connections and YouTube for sharing educational contents). Studying could differentiate the impact of every platform on students' academic involvement and performance. This would enable the educators to recommend tools in platforms tailored towards academic usage, hence assist the students to use those appropriate for the course.

5.5.3. Physical and Mental Health Impacts

This paper therefore aims at exploring the relationship between the duration spent on social media and effects on health. Researchers could perform further studies in the future to understand how social media affects visual health by causing eye strain while staying indoors and how their stressful academic environment affects their mental health.

➤ REFERENCES

1. Chugh, R., Ruhi, U., & Joshi, M. (2021). Social media in higher education: A framework for continuous engagement. *Education and Information Technologies*, 26(6), 7477–7496. <https://doi.org/10.1007/s10639-021-10804-9>
2. Deng, L., & Tavares, N. J. (2013). Understanding students' perceptions on the benefits of online social networking use for teaching and learning. *The Internet and Higher Education*, 17, 39–45. <https://findanexpert.unimelb.edu.au/scholarlywork/959132>
3. Manca, S., & Ranieri, M. (2013). Identifying the use of online social networking in higher education. *Educational Technology & Society*, 16(1), 107–118. Retrieved from https://www.researchgate.net/publication/255639900_Identifying_the_use_of_online_social_networking_in_higher_education
4. Ansari, J. A., & Khan, N. A. (2020). Exploring the role of social media in collaborative learning: The new domain of learning. *Smart Learning Environments*, 7:9. [<https://doi.org/10.1186/s40561-020-00118-7>]
5. Moran, M., Seaman, J., & TintiKane, H. (2011). Teaching, learning, and sharing: How today's higher education faculty use social media. Babson Survey Research Group. [https://www.researchgate.net/publication/247936322_]
6. Gikas, J., & Grant, M. M. (2013). Mobile computing devices in higher education: Student perspectives on learning with social media. *Internet and Higher Education*, 19, 18–26. [\[https://doi.org/10.1016/j.iheduc.2013.06.002\]](https://doi.org/10.1016/j.iheduc.2013.06.002)
7. Greenhow, C. (2011). Online social networks and learning. *On the Horizon*, 19(1), 4–12. [\[https://doi.org/10.1108/10748120110424816\]](https://doi.org/10.1108/10748120110424816)
8. Azizi, A., Soroush, A., & Khatony, A. (2019). The relationship between social networking addiction and academic performance in Iranian students. *BMC Psychology*, 7(28). <https://doi.org/10.1186/s40359-019-0302-1>
9. Sharmin, N., Ahmed, O., & Ahmed, Z. (2019). Role of Facebook use and addiction on academic achievement: A pilot study on undergraduate students. *Bulgarian Journal of Science and Education Policy*, 13(1), 92–108.
10. Nasrullah, S., & Khan, M. S. (2015). The impact of time management on students' academic achievement. *Journal of Literature, Languages and Linguistic*, 2, 66–71. Retrieved June 15, 2020, from https://www.researchgate.net/publication/313768789_The_Impact_of_Time_Management/
11. Rouse, M. (2019). Time management. Retrieved from <https://whais.techtarget.com/definition/time-management>
12. Goodluck, E. (2018). 8 steps to effective time management for students. Retrieved from <http://www.goodluckexams.com/8-steps-to-effective-time-management-for-students/>
13. Ahmed, N. (2019). Statistical analysis of impacts of peer pressure and time management on the academic performance of students. Retrieved from <https://steemit.com/stemng/@rubies/statistical-analysis>

ANNEXURES

- a. What is your age?
- b. Which department or field of study are you currently enrolled in?
- c. What year of study are you in?
- d. How often do you use social media for academic purposes?
- e. How long have you been using social media as a tool for academic learning?
- f. Which social media platform do you find most effective for learning-related activities?
- g. How helpful do you find academic resources shared on social media for your studies?
- h. To what extent has social media improved your ability to collaborate with peers on academic projects?
- i. Overall, how beneficial do you find social media as a learning tool in supporting your academic progress?
- j. How often do you feel distracted by non-academic content while using social media for study purposes?
- k. How often do you feel overwhelmed by the amount of information available on social media when looking for academic content?
- l. How does using social media for academic purposes affect your focus and productivity?
- m. How many hours per day do you typically spend on social media for academic purposes?
- n. Has using social media for learning purposes affected your sleep patterns?
- o. Do you experience any physical discomfort related to using social media for academic purposes?
- p. Has the time you spend on social media for learning purposes reduced the time you dedicate to physical activity or exercise?